

Planning under uncertainty: delivery outline (v2)

UNC Computational Medicine Seminar, September 4, 2026, 45 minutes

Three sections, one argument; 43 main slides + 11 backups. Total advances about 80. Reach the program-failure slide (37) with ~10 minutes left; slides 37 to 41 get 6 to 7 protected minutes.

Spoken pivots: Section 1 to 2: *“The problem was never simulation. It was what to simulate next.”* Method to philosophy (slide 25): *“Which of the feasible designs should we optimize for?”* Section 2 to 3: *“All of this is machinery unless it earns trust and changes the actual trial.”*

The main deck, in order

#	Slide	Adv	The one thing to land
1	Title		Symposium-style title, transparent hex, logo strip. No lab intro.
2	Overview		30 to 45 seconds spoken over the bullets; section 3’s second line is a promise, not a spoiler.
3	Section 1: The problem		Divider; optional spoken hook already lives on the next slide’s notes.
4	ADAPT challenge		ARPA-H asked for biomarkers, treatments, trials co-evolving. One sentence, fast.
5	Future state		Official ARPA-H longitudinal schematic: measure before, during, after; therapies switch. One sentence.
6	ADAPT components		Three integrated components; our team leads the clinical-trial one.
7	UNC team + EVOLVE		Full org chart, steering committee and stat working group outlined.
8	What ARPA-H asked	3	Excerpts, new-ask, the ask, meme. Let each be read.
9	Collision		The right column is the rest of the talk. 30 s.
10	One equation	1	For a simple fixed design: one equation, one unknown. Hold that thought.
11	Bayesian adaptive knobs	1	Timelines, then: every interim decision rule is a setting; 8 to 13 interacting choices.
12	Count them	1	FDA targets up top; null box (orange) and alternative box (blue); feasible = targets met. Then THE pivot.
13	Two weeks	1	First person. Power stalled at 0.78; more subtrials waiting.
14	Alternatives fail	2	Grid panel, then random panel: $136/2,000 = 6.8\%$. Manual tuning already covered on 13.
15	10-15 subtrials		Section 1 escalation: not one problem, 10 to 15, on a short clock.
16	Not unique	1	First person: phase I and II hit the same wall. Same shape every time.
17	Section 2: BATON		Speak the pivot: the problem was never simulation.
18	ML bridge	2	ML side first, then trial side, then the constraint difference.
19	BATON + loop	2	Card first, then the loop, then the landing.
20	Initial sample		BATON step 1: space-filling seed, no model yet. One beat.

#	Slide	Adv	The one thing to land
21	Surrogate map	2	Points first, then the fitted curve with its band: settings in, operating characteristics out.
22	Acquisition		The orange point is not necessarily at the predicted best.
23	Seed vs walk	2	The payoff: seed sketches the map, model places every evaluation after.
24	Fidelity ladder		Explore at 3k, validate top designs at 10k, escalate only if needed, verify at 5 seeds. Never say “three rounds”.
25	Hinge	1	Finding feasible designs is not the whole problem. Slow down.
26	Two designs	1	Different objectives; neither universally better.
27	Philosophies	4	Merged: feasibility is not preference; three philosophy cards; transport claim.
28	Section 3: Evidence		Speak the pivot: does it earn trust, and did it matter?
29	Simon unit test		10 random starts: 8 return the exact Simon design, 2 near-neighbors 0.15 away. Never “recovers”.
30	Benchmark		12 scenarios: 12/12 feasible, 4/12 exact, 2.5% median, 13.5% worst.
31	Frontier: single-arm	1	EVOLVE’s TNBC cohort; speak the randomized price line (B9).
32	Seamless		13 interacting parameters make manual calibration impractical.
33	Collapse		The identical design. Label now clear of the frontier line.
34	Gate		Stage 1 dominates under the null; objectives align. Search as instrument.
35	Operational impact	5	Boxes, then the Admissible line. Bridge moves to 36.
36	JASA + software		Cropped title page + hex, 20 s, then SPEAK THE BRIDGE here: is the trial the right unit?
37	Program failure	3	Hero numbers: 54% late-stage; 21 of 138 dose PMRs. Well designed locally, poorly optimized globally.
38	Sotorasib	1	960 mg carried to approval; FDA-required 240 vs 960; similar PFS at one-quarter the dose.
39	Extensions	3	Toward phaseless development: possible futures, earlier decisions, across trials. Goal line.
40	Development programs	3	Pipeline dims, reversal chain, then huge: phase structure becomes a design variable. 2+ min.
41	Summary	3	Three lines; spoken close on which workable design we want, and why.
42	Acknowledgments		MPIs and statisticians by name; funding is the footer row.
43	Questions		Leave up; routing index in notes.

Backups (44 to 54)

B1 GP/acquisition (code-verified answers) · B2 Simon per-scenario · B6 manual-vs-BATON in full · B3 parameters and assumed truths · B5 related work · B4 bidirectional · B10 known failure modes and limits · B7 what I am not arguing · B8 ten-seed reproducibility · B9 randomized-cohort frontier.

If running long

Planned cuts already made. Remaining option: slide 9 (collision table), one spoken sentence. Emergency valve: past 30 minutes at the Section 3 divider, speak slide 29’s headline numbers while advancing to the EVOLVE frontier.

Q&A quick routes

Conventions: slide 10 notes. Bayesian-frequentist hybrid: slide 11 notes. Random search “beat you”: slide 14 notes, then 29 to 30. GP/acquisition/CRN detail: backup B1. Simon seed accounting: slide 29 notes, B2. Collapse artifact: slide 33 notes. Unspent alpha: slide 31 notes. Multi-objective BO, philosophy lock: slide 27 notes. Two-week detail: B6. Company/translation: one sentence, disclose verbally.